503. Next Greater Element 2:

Given a circular integer array `nums` (i.e., the next element of `nums[nums.length - 1]` is `nums[0]`), return *the **next greater number** for every element in* `nums`.

The **next greater number** of a number `x` is the first greater number to its traversing-order next in the array, which means you could search circularly to find its next greater number. If it doesn't exist, return `-1` for this number.

Example 1:

```
Input: nums = [1,2,1]
Output: [2,-1,2]
Explanation: The first 1's next greater number is 2;
The number 2 can't find next greater number.
The second 1's next greater number needs to search circularly, which is also 2.
```

Example 2:

```
Input: nums = [1,2,3,4,3]
Output: [2,3,4,-1,4]
```

Constraints:

- `1 <= nums.length <= 104`
- `109 <= nums[i] <= 109`

Solution: time = O(N*N) and space = O(n)

```
class Solution:
    def nextGreaterElements(self, nums: List[int]) -> List[int]:
        n = len(nums)
        ans = []
        nums.extend(nums)
```

```python
        l = 0
        while l < n:
            r = l+1
            while r < len(nums):
                if nums[r] > nums[l]:
                    ans.append(nums[r])
                    break
                else:
                    r += 1
            if r == len(nums):
                ans.append(-1)
            l += 1
        return ans
```